

Lubricant is defined as a substance which reduces the friction between two moving metallic surfaces, and the phenomena known as lubrication.

* Functions of Lubricants

- 1) To reduce friction
- 2) Reduce deformation & wear & tear.
- 3) It work as a coolant
- 4) It work as a seal.
- 5) It prevent corrosion.
- 6) It increase the efficiency of machine.

* Classification of Lubricants

① Liquid Lubricants

It includes animal oils, vegetable oils, petroleum oils, synthetic lubricants.

Animal oils - Tallow oil, whale oil etc.

Vegetable oil - Castor oil, palm oil etc.

Petroleum oil - Petroleum fractions.

Synthetic lubricants - Polyglycol, silicones etc.

② Semi-Solid Lubricants :-

Semi-solid lubricants are formed by emulsifying oil & fat with thickening agents like soap of sodium, calcium, lithium, aluminum

at higher temperature. eg. $\rightarrow$ Grease.

Classification :-

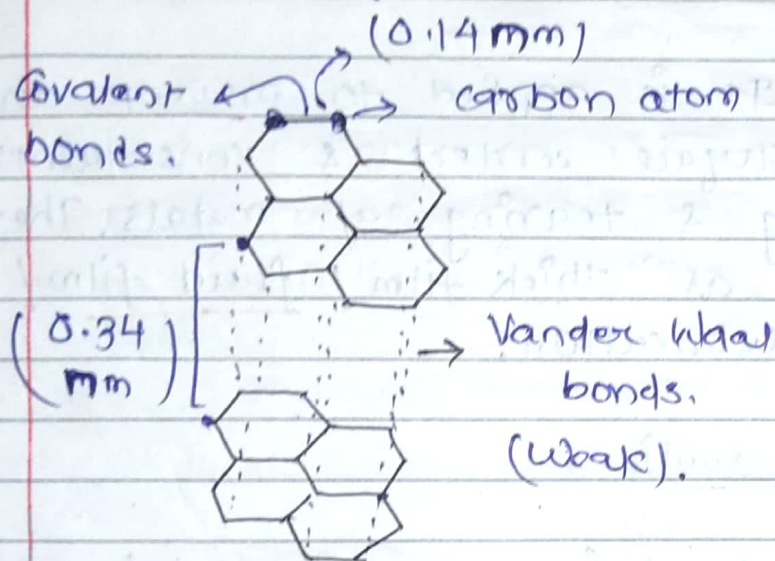
- i) Soda based — In this case sodium soap are used as a thickening agent in mineral or petroleum oil. They are slightly soluble in water. They can be used up to 175°C .
- ii) Lithium based : — In this case Li soap are emulsifying with petroleum oil. They are water resistant and used up to 15°C .
- iii) Calcium based : — In this case Ca-soap are emulsifying with petroleum oil. They are also water resistant and used up to 80°C . At higher temp. soap & petroleum oil are separate from each other.

③ Solid lubricants :-

Graphite, Molybdenum disulphide (MoS_2), boron nitride (BN), are predominantly used as a solid lubricants. They are used under high temp. & high load (pressure).

eg. (i) Graphite :- It is most widely used as a solid lubricant. Graphite has layer structure, layers are held together.

with the help of weak Vander Waal's forces which facilitate the easy sliding of one layer on the other layer. It is very soapy to touch, non-inflammable. It is used at higher temp (around 450°C). They are either used as powder form or mixed with oil or water.



- (ii) MoS_2 :- It is sandwich like structure in which hexagonal layers of Mo lies between two hexagonal layers of Sulfur (S) atom. Like graphite each layers are held together with weak Vander Waal's forces. It is stable up to 400°C . It is differ from graphite. because it is used in high vacuum unlike graphite. It adheres even more strongly to the metal or other surface.

* Mechanism of Lubrication

① Thick film / Fluid film / Hydrodynamic Lubrication

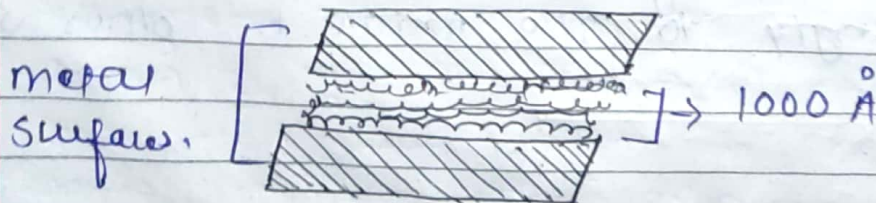
It is carried out with the help of Liquid lubricants. In this ~~mechanism~~ mechanism, two moving & sliding surfaces are separated by thick film of lubricants fluid of about 1000 Å.

It is applied to prevent direct surface to surface contact & consequently reduce wearing & tearing of metals. Therefore it is known as thick film / fluid film / Hydrodynamic lubrication.

↓ ↓
Liquid Relative motion.

In this case fluid is formed by mixing of hydrocarbon oils & anti-oxidants with long chain polymer so as to maintain viscosity.

eg. It is useful in delicate & light machines like, watches, clocks, guns, scientific equipments, (high speed, low load).

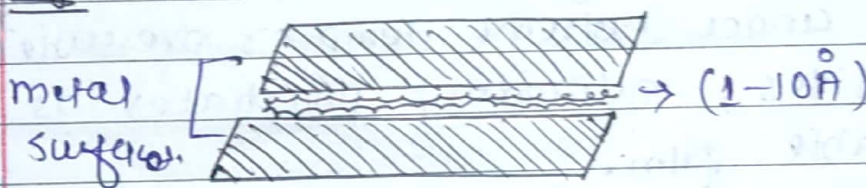


② Thin film / Boundary Lubrication :-

It is carried out with semi-solid (grease), and solid (graphite / MoS₂) lubricants.

Boundary lubrication is a condition in which the lubricant film becomes too thin to provide total separation, ($1-10 \text{ \AA}$). In this type of lubrication a thin film of lubricant is adsorbed on the surface by weak Van der Waal's forces.

Thin film lubrication is operating at relatively low speed & heavy loading (pressure) eg. semi-solid lubricants (Grease), solid lubricants.



③ Extreme Pressure / Temperature Lubrication.

~~It is carried out with semi-solid and solid lubricants. Boundary lubrication is a condition in which the lubricant film becomes too thin to provide total~~

In this mechanism, moving & sliding surfaces are under high pressure & speed, therefore this mechanism is known as extreme pressure lubrication.

~~eg. semi-solid lubricants (Grease).~~

In such a case high temp. generated due to friction, under these condition liquid lubricants are fail to stick and decompose or vapourize. These problems are minimized by special additives are added to minerals oils.

These additives form durable films on material (metal surfaces) which can withstand high load & high temp.

Important additives are organic compound having group like chloride, sulphur, phosphorus etc. They react with metallic surface to form metallic compound (posses high melting point & serve as good lubricants under extreme temp or pressure). like chlorides, sulphides, phosphates as more durable film.

eg. chlorinated esters, sulphurized oil

	Thick film	Thin film	Extreme pressure.
1)	High speed Low pressure.	Low speed High pressure	High speed High pressure High temp.
2)	1000 Å	1-10 Å	—
3)	Liquid lubricants	Semi-solid / Solid lubricants	Solid lubricants
	No direct contact b/w metal surfaces	Direct contact b/w metal Surfaces.	—

* Properties of Lubricants

1) Viscosity :-

Viscosity is the property of a fluid that determines its resistance to flow. It is an Indicator of flow ability of lubricating oil. If temperature increases viscosity of the lubricating oil decreases & pressure increases viscosity of lubricating oil increases.

In short we can say that good lubricating oil is that whose viscosity does not change with temperature.

2) Viscosity Index :- The variation of viscosity of a liquid with temperature is called viscosity index. A relatively small change in viscosity with temp. is indicated by high viscosity oil (Pennsylvania oil) index whereas, a low viscosity index shows, a relatively large change in viscosity with temp. indicated by low viscosity index (Gulf oil).

→ (Gulf oil)
low viscosity index oil (at 100°F).

$$VI = \frac{L - U}{L - H} \times 100$$

Viscosity of test oil at 100°F

↓ high viscosity index oil (at 100°F)
(Pennsylvania oil)

3) Flash point :- The flash point of a volatile material is the lowest temp. at which vapours of the material will ignite for a moment when an ignition source brought near to it.

The lubricating oil should have flash point reasonably above its working temperature.

4) Fire point :- The fire point of a fuel is the lowest temperature at which the vapour of that fuel will continue to burn at least 5 sec. when an ignition source brought near to it. Fire point is around 10°C higher than flash point.

5) Cloud point :-

When we cool down the oil, the temperature at which lubricating oil becomes cloudy in appearance is called cloud point.

6) Pour point :- The lowest temperature at which the lubricating oil become semi-solid and ceases to flow is called pour point. It indicates the suitability of lubricants used in cold condition. Good lubricant should possess low pour point.

Flash point → Can be determined by Pensky marten apparatus & Able's Apparatus.

Cloud point
Power point → Can be determined by cloud & power point apparatus.

7) Steam Emulsification No :-

Emulsion → Liquid + Liquid

(oil + water) ⇒ Not good.

[attracts dirt particle, It causes wear & tear.]

The time in seconds in which oil & water emulsion separates out in distinct layers,

Unit → [Second]

For a good lubricating oil SEN must be low